

Digital Filter Theory & Synthesis Equations

Mathematical prototypes, bilinear frequency pre-warping, and cascaded Direct Form II Transposed biquad sections.

Biquad Second-Order Sections (SOS) Cascade

Higher-order digital IIR filters are synthesized as a series cascade of $K = \lceil N/2 \rceil$ Second-Order Sections to guarantee numerical robustness and suppress coefficient quantization drift:

$$H(z) = g \cdot \prod_{k=1}^K \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (1)$$

where g represents composite scale gain. In Overtune 3, each section executes the **Direct Form II Transposed** topology:

Difference Equations:

$$y_k[n] = b_{0k}x_k[n] + w_{1k}[n-1] \quad (2)$$

$$w_{1k}[n] = b_{1k}x_k[n] - a_{1k}y_k[n] + w_{2k}[n-1] \quad (3)$$

$$w_{2k}[n] = b_{2k}x_k[n] - a_{2k}y_k[n] \quad (4)$$

A cascade is strictly BIBO stable if and only if all poles reside inside the unit circle: $|p_{ik}| < 1.0 \ \forall i, k$.

Analog Prototype Approximations

1. Butterworth (Maximally Flat)

Optimized for zero passband ripple and monotonic attenuation:

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad (5)$$

Closed-form stable left-half plane poles distributed on a circle of radius Ω_c :

$$s_k = \Omega_c \exp\left(j \left[\frac{(2k + N - 1)\pi}{2N} \right]\right), \quad k = 1, 2, \dots, N \quad (6)$$

2. Chebyshev Type I (Passband Equiripple)

Sharper transition roll-off achieved by trading off passband equiripple variation ϵ :

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2\left(\frac{\Omega}{\Omega_c}\right)}, \quad \epsilon = \sqrt{10^{\delta_{\text{pass}}/10} - 1} \quad (7)$$

where $T_N(\omega) = \cos(N \arccos \omega)$ for $|\omega| \leq 1$. Pole locations lie on an ellipse:

$$s_k = -\Omega_c \sinh(\gamma) \sin \theta_k + j \Omega_c \cosh(\gamma) \cos \theta_k, \quad \gamma = \frac{1}{N} \operatorname{arsinh}(1/\epsilon), \quad \theta_k = \frac{(2k-1)\pi}{2N} \quad (8)$$

3. Chebyshev Type II (Inverse Chebyshev)

Maximally flat passband with equiripple stopband attenuation δ_{stop} (dB) and pure imaginary transmission zeros:

$$|H(j\Omega)|^2 = \frac{1}{1 + \frac{1}{\epsilon_s^2 T_N^2(\Omega_s/\Omega)}}, \quad z_m = j \frac{\Omega_s}{\cos\left(\frac{(2m-1)\pi}{2N}\right)}, \quad m = 1, \dots, \lfloor N/2 \rfloor \quad (9)$$

4. Elliptic / Cauer (Optimal Transition Sharpness)

Yields the steepest transition roll-off for a given filter order by permitting equiripple behavior in both pass-band and stopband:

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 R_N^2(\xi, \Omega/\Omega_c)} \quad (10)$$

where $R_N(\xi, \omega)$ is the Chebyshev rational function and $\xi \geq 1$ is the selectivity factor. Zeros are placed on the imaginary axis to generate deep attenuation nulls.

5. Bessel-Thomson (Linear Phase & Zero Transient Ringing)

Engineered to preserve audio transients, square waves, and percussion impulses without ringing or overshoot:

$$H(s) = \frac{\theta_N(0)}{\theta_N(s/\omega_0)}, \quad \theta_n(s) = (2n-1)\theta_{n-1}(s) + s^2\theta_{n-2}(s), \quad \theta_0 = 1, \quad \theta_1 = s + 1 \quad (11)$$

Group delay $\tau_g(\omega) = -d\phi(\omega)/d\omega$ is maximally flat around $\omega = 0$.

Bilinear Transform & Tangent Frequency Pre-Warping

The continuous-time prototype $H(s)$ maps algebraically to discrete-time $H(z)$ via trapezoidal integration:

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}}, \quad T_s = \frac{1}{F_s} \quad (12)$$

To cancel frequency warping near the Nyquist limit $F_s/2$, analog cutoff frequencies are pre-warped prior to substitution:

Pre-Warping Formula:

$$\Omega = 2F_s \tan\left(\frac{\pi F_c}{F_s}\right) \quad (13)$$

For bandpass and notch configurations, lower (F_{c1}) and upper (F_{c2}) edges are pre-warped independently, preserving geometric center frequency $\omega_0 = \sqrt{\omega_1\omega_2}$ and bandwidth $B = \omega_2 - \omega_1$.

Summary of Prototype Characteristics

Filter Family	Passband	Stopband	Phase Response	Primary Audio Use Case
Butterworth	Flat	Monotonic	Non-linear	General audio mastering, crossovers
Chebyshev I	Equiripple	Monotonic	Moderate dispersion	Steep sub-bass cuts, band isolation
Chebyshev II	Flat	Equiripple	Moderate dispersion	Anti-hum notch filtering
Elliptic	Equiripple	Equiripple	Highly non-linear	Brickwall anti-aliasing
Bessel	Drooping	Monotonic	Linear (Constant delay)	Mastering drums, transients

External References

- [Butterworth Filter Theory – Wikipedia](#)
- [Chebyshev Filters \(Type I & II\) – Wikipedia](#)
- [Elliptic \(Cauer\) Rational Filters – Wikipedia](#)
- [Bessel-Thomson Linear Phase Filters – Wikipedia](#)
- [Bilinear Transform with Pre-Warping – Wikipedia](#)
- [Digital Biquad Second-Order Sections \(SOS\) – Wikipedia](#)
- [Overtune 3 Project Source Code Repository](#)